MAMI Human Milk Microbiome — Next-stage quality and association analysis

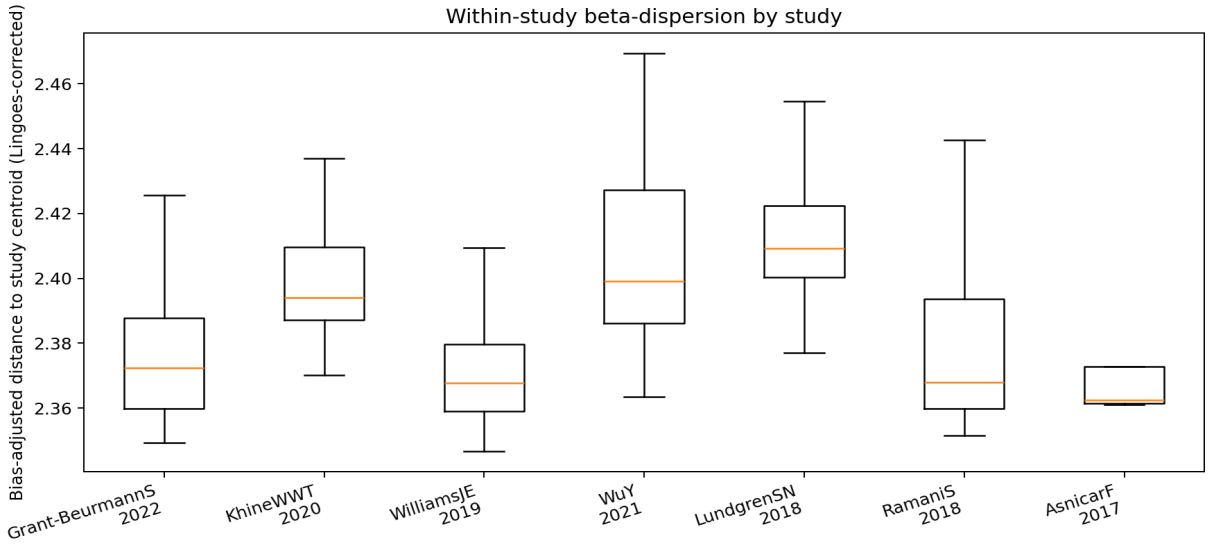
Taxonomy-name review, within-study dispersion, prevalence sensitivity and exploratory feature associations

Data scope. The analysis uses 1,128 MAMI breast-milk samples, linked by Run ID to study metadata. Of 580 original feature columns, 346 were non-zero at least once. The working base set excludes only three explicit host/organelle labels (Mammalia, Chloroplast and Mitochondria), leaving 344 features. Samples were re-closed to relative abundances after this filtering.

1. Beta-dispersion diagnostic

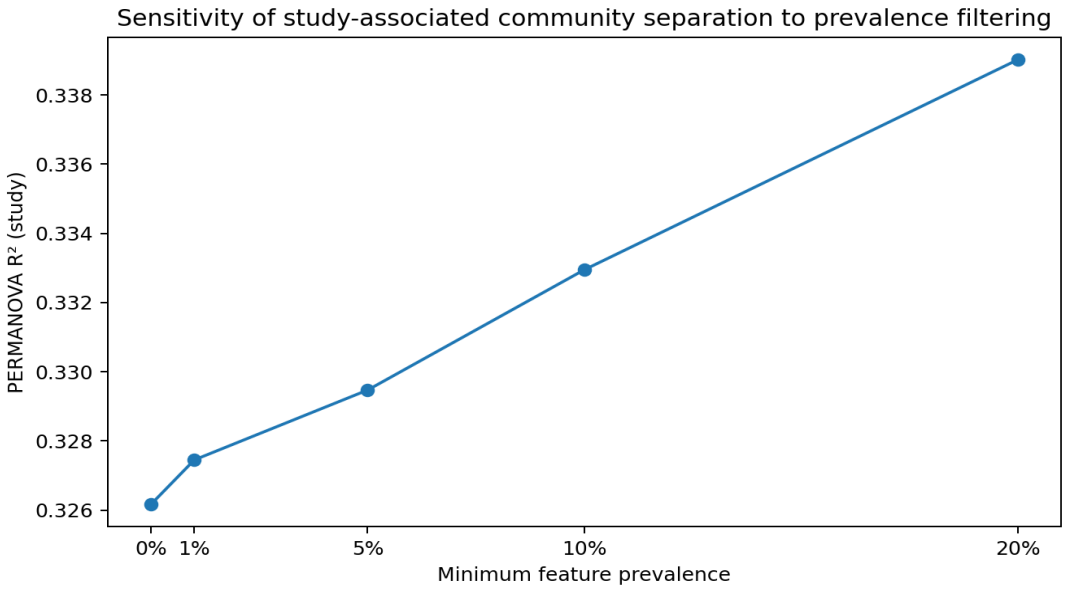
The study-level PERMANOVA from the prior stage was pseudo-F = 90.43, R^2 = 0.326, p = 0.001 (999 permutations). The PERMDISP-style test was also significant: F = 36.94, p = 0.001 (999 permutations). It used Bray–Curtis distances, a Lingoes correction for non-Euclidean distances, and a small-sample bias adjustment for unequal group sizes.

This means unequal within-study spread is present as well as study-associated community separation. In particular, the group sizes are very unbalanced (one study has only five samples), so the PERMANOVA result should not be interpreted as a pure difference in group centroids or as evidence of country effects.



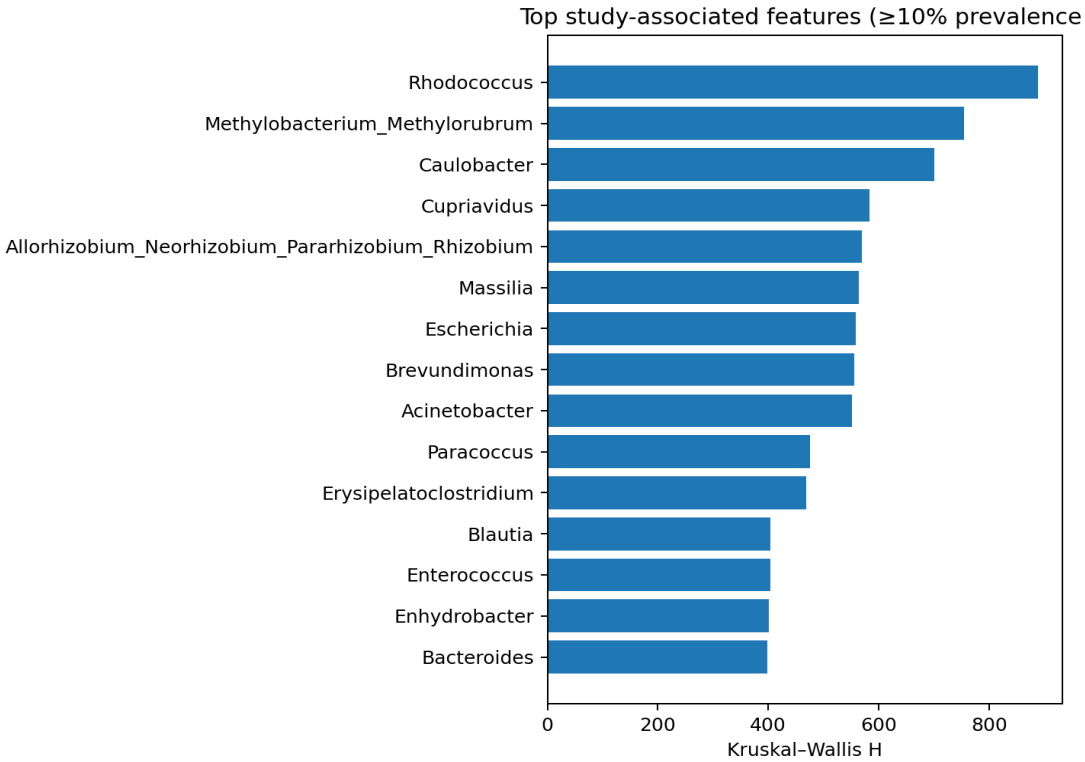
2. Prevalence-filter sensitivity

The PERMANOVA was repeated after retaining features present in at least 0%, 1%, 5%, 10% and 20% of samples. R^2 ranged from 0.326 to 0.339 and increased modestly as rare features were removed. This suggests that the study-associated signal is not being driven solely by very rare features. Each sensitivity PERMANOVA used 199 permutations; the p-values are therefore coarse (minimum reportable p = 0.005).



3. Exploratory feature-level screen

For features present in at least 10% of samples, Kruskal–Wallis tests compared relative-abundance distributions across the seven studies; Benjamini–Hochberg correction was applied. 54 of 54 tested features had q < 0.05. The most strongly study-associated feature labels included Rhodococcus, Methylobacterium_Methylorubrum, Caulobacter, Cupriavidus and Allorhizobium_Neorhizobium_Pararhizobium_Rhizobium.



Feature	Prevalence	Highest median in	BH q-value
Rhodococcus	50.8%	Grant-BeurmannS_2022	6.1e-187
Methylobacterium_Methylorubrum	63.1%	Grant-BeurmannS_2022	2e-158
Caulobacter	11.1%	LundgrenSN_2018	5.3e-147
Cupriavidus	41.7%	Grant-BeurmannS_2022	1.5e-121
Allorhizobium_Neorhizobium_Pararhizobium_Rhizobium	16.7%	LundgrenSN_2018	6.3e-119
Massilia	12.9%	LundgrenSN_2018	1.4e-117

Feature	Prevalence	Highest median in	BH q-value
Escherichia	24.9%	WuY_2021	1.6e-116
Brevundimonas	21.9%	LundgrenSN_2018	6.6e-116
Acinetobacter	56.7%	WuY_2021	3.4e-115
Paracoccus	20.8%	RamaniS_2018	1e-98

These are study-associated feature signals, not validated differential-abundance biomarkers. The tests use relative abundances, and the study grouping is entangled with geography, sequencing protocols and other study-specific factors. They should not be called country-specific microbial markers.

4. Taxonomy and contamination review

The uploaded feature table provides taxon-like labels, but no reference sequences, full taxonomy lineages, classifier/database versions, raw reads or negative-control profiles. Taxonomy therefore cannot be independently verified and contaminants cannot be confidently identified from these files alone. The review CSV flags obvious host/organelle labels, one detected archaeal feature, unresolved group labels, and genera for manual comparison against negative controls. No genus was automatically removed as a presumed contaminant.

This caution is particularly relevant because human milk is a low-biomass sample type. Human-milk-specific methods papers describe contamination from environmental, reagent and experimental sources as a major challenge and recommend positive and negative controls. General low-biomass guidelines likewise recommend sequencing controls alongside biological samples.

5. What this means for the project

Supported: the supplied MAMI matrix shows strong study-associated differences and a set of features whose relative abundances vary across studies. **Not supported:** a causal country effect, a clinical association, proof that flagged genera are contaminants, or a claim that specific features are biologically enriched in a country.

Before manuscript-grade differential-abundance analysis, the essential missing inputs are raw counts (or sequence reads), full taxonomy, study-specific processing details, and negative-control results. If these cannot be obtained, conclusions must remain exploratory and explicitly study-associated.

Exploratory computational analysis only; not a clinical finding or diagnostic tool.

Selected methodological references: Tobin et al. (2018), doi:10.3791/56974; Gacesa et al. (2023), doi:10.1038/s43705-023-00325-6; Eisenhofer et al. (2019), doi:10.1016/j.tim.2018.11.003; Salter et al. (2014), doi:10.1186/s12915-014-0087-z; Nature Microbiology low-biomass guidelines (2025), doi:10.1038/s41564-025-02035-2.